Reg. No. | | | | | | | | | | | [14AE71]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2018

AEROSPACE CONTROL SYSTEMS

Time:3 Hours **Maximum Marks:70**

Instructions:
- **i.** Answer any **FIVE** full Questions choosing atleast two from each part.
- **ii.** Assume missing data (if any) and state the same.
- **iii.** Use semi log graph sheet and ordinary graph sheet as required.

PART - A

1. a. What are the effects of feedback on control system performance? Discuss. **04**

b. Compare linear and nonlinear systems. **04**

c. Find the inverse Laplace transform of the following transfer function **06**

$$F(s) = \frac{5(s+2)}{s^2(s+1)(s+3)}$$

2. a. Obtain the transfer function Y(s)/U(s) of the system shown in Figure 2(a). The input u is a displacement input. **11**

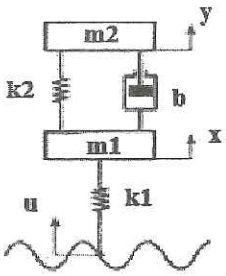

Fig. 2(a)

b. Describe basic building blocks associated with electrical circuits. **03**

3. a. Describe, with graphical, mathematical representations in time domain and in Laplace domain, any two types of inputs normally used for studying the response characteristics of linear control systems in time domain. **04**

b. A UFBCS is characterized by an OLTF $G(s) = \frac{10}{s^2 + 5s + 6}$ Determine the following, when the system is subjected to unit step input: **10**

- **i.** Undamped natural frequency,
- **ii.** Damping ratio,
- **iii.** Peak overshoot,
- **iv.** Peak time and
- **v.** Settling time.

4. a. Write a technical note on higher order systems. **06**

b. Show that the satellite attitude control system shown in Figure 4(a) exhibits continued oscillations, and find its frequency of oscillation if K / J = 4. Now, show that the system can be stabilized by using tachometer feedback, as shown in Figure 4 (b) and find the value of K_h that will yield the damping ratio to be 0.6. **08**

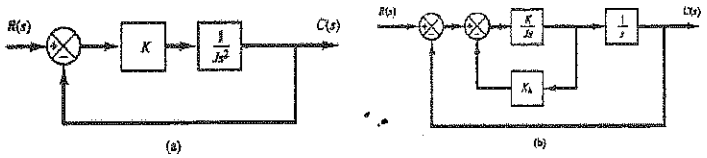

Fig. 4

PART - B

5. a. Plot the root loci for the system with transfer function shown in Fig (5), on a graph sheet, showing all the calculations and mark all salient points on the graph. Calculate and plot on the graph, the roots corresponding to K = 5 **14**

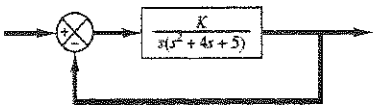

Fig. (5)

6. a. Given $G(s) = \omega_n^2 / (s^2 + 2\varsigma\omega_n s + \omega_n^2)$ show that $|G(j\omega_n)| = \frac{1}{2\varsigma}$ **04**

b. Draw a Bode diagram of the open-loop transfer function G(s) given in the Fig. 6(b). Determine the phase margin and gain margin. **10**

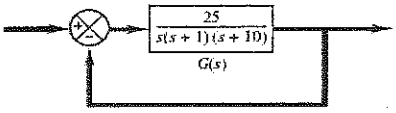

Fig 6(b)

7. Design a compensator such that the dominant closed-loop poles are located at s = -1 ± j1 for the system shown in Fig (7), using root locus approach. **14**

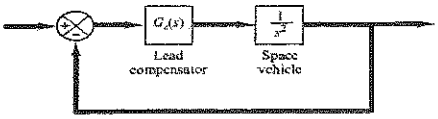

Fig. (7)

8. a. Define (i) State; (ii) State Variable; (iii) State trajectory **04**

b. State the properties of state transition matrix Φ(t). **04**

c. Derive the state and output model of the mechanical system shown in Fig 8(c) and express in vector matrix form. **06**

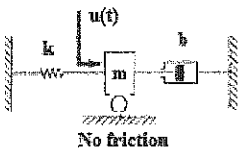

Fig 8(c)